FAST-NUCES

Department of Artificial Intelligence & Data Science

CS-4112: Deep Learning

Assignment 2: Reproducibility Study

Denoising Diffusion Probabilistic Models on CIFAR-10

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Due Date: 5 April 2026

1. Introduction

This report presents a reproducibility study of Denoising Diffusion Probabilistic Models (DDPM) as introduced by Ho et al. (2020), a foundational work in deep generative learning. The study was conducted as part of Assignment 2 for CS-4112 Deep Learning at FAST-NUCES. The primary objective was to reproduce the core experimental results of the paper using the same dataset (CIFAR-10) and evaluation metric (Fréchet Inception Distance, FID), and to critically analyze all differences between our reproduced results and the paper-reported values.

Diffusion models learn to synthesize realistic images by progressively reversing a Gaussian noise corruption process applied over hundreds of timesteps. The DDPM framework established a clean mean-squared-error training objective and demonstrated that diffusion models could match or exceed GANs on image quality benchmarks without adversarial training. On CIFAR-10, Ho et al. reported FID of 3.17 computed on 50,000 generated samples, which became a widely referenced baseline for subsequent generative modeling research including Consistency Models (Song et al., ICML 2023) and EDM (Karras et al., NeurIPS 2022).

In this study, we reproduce the DDPM inference pipeline using the official pretrained CIFAR-10 model published through the HuggingFace `diffusers` library. We evaluate image generation under three sampling configurations: standard DDPM with 1000 denoising steps, accelerated DDIM sampling with 10 steps, and single-step DDIM sampling. We compute FID for each configuration using 5,000 generated images, compare against paper-reported values, and discuss all sources of discrepancy in detail.

We also attempted to reproduce results from two baseline papers: Consistency Models (Song et al., ICML 2023) and EDM (Karras et al., NeurIPS 2022). The official Consistency Models checkpoint was found to be permanently deleted from OpenAI's servers and HuggingFace, which constitutes a significant reproducibility barrier documented in Section 6. EDM baseline numbers are cited from the paper for contextual comparison.

2. Paper Overview

2.1 Primary Paper: Denoising Diffusion Probabilistic Models (Ho et al., NeurIPS 2020)

The paper introduces Denoising Diffusion Probabilistic Models, a class of latent variable generative models that reverse a fixed Markov chain of Gaussian noise additions. The framework consists of two processes: a fixed forward process that corrupts data, and a learned reverse process that reconstructs clean data from noise.

Problem: Given samples from an unknown data distribution (e.g., CIFAR-10 images), learn a model that generates new samples from the same distribution without adversarial training.

Forward Process: A fixed Markov chain adds Gaussian noise at each of $T = 1000$ timesteps according to a variance schedule. After 1000 steps, any image becomes indistinguishable from standard Gaussian noise.

Reverse Process: A deep U-Net neural network with encoder-decoder structure and

skip connections is trained to predict the noise added at each timestep given the noisy image and timestep index. During inference, the model iteratively denoises from pure Gaussian noise over 1000 steps to produce a realistic image.

Training Objective: A simplified mean-squared-error loss ($\mathcal{L}_{\text{simple}}$) between the network's predicted noise and the actual noise added. This is numerically stable and requires no adversarial training, a major advantage over GANs.

CIFAR-10 Results: FID of 3.17 and Inception Score of 9.46, computed using 50,000 generated samples compared against all 50,000 CIFAR-10 training images as the real reference set.

2.2 Baseline Paper 1: Consistency Models (Song et al., ICML 2023)

Consistency Models address the fundamental sampling inefficiency of DDPM: generating one image requires 1000 sequential neural network evaluations. The key contribution is a self-consistency constraint — any point along a diffusion trajectory, when passed through the consistency model, should map directly to the same clean image endpoint. This enables generation in as few as one or two network evaluations.

The paper presents two training approaches: consistency distillation (distilling from a pretrained DDPM) and consistency training (training from scratch without a pretrained diffusion model). On CIFAR-10, consistency distillation achieves FID of 3.55 with a single sampling step and 2.93 with two steps — nearly matching DDPM quality at hundreds of times the speed. This paper was published at ICML 2023 (arXiv:2303.01469) by Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Our DDIM experiments directly demonstrate why specialized training is needed: standard DDPM sampling with only 1 step produces pure noise (FID 462), validating the problem that Consistency Models solve.

2.3 Baseline Paper 2: EDM (Karras et al., NeurIPS 2022)

Elucidating the Design Space of Diffusion-Based Generative Models (EDM) provides a clean, modular framework that disentangles the key design choices in diffusion models, including noise schedules, network preconditioning, loss weighting, and sampling procedures. By analyzing the probability flow ODE and introducing improvements such as a higher-order Heun sampler, better time discretization, and principled preconditioning of the score network, EDM achieves state-of-the-art FID scores on CIFAR-10 of 1.97 (unconditional) and 1.79 (class-conditional) using only 35 network evaluations per image.

The paper demonstrates that careful attention to training and sampling dynamics can yield large quality gains without altering the core U-Net architecture. This is significant because it shows that the gap between our reproduced DDPM FID (20.91) and the paper's original 3.17 is not purely a sample-size issue — there is genuine room for improvement through better training methodology. Both EDM and Consistency Models build directly on the DDPM foundation reproduced in this study, making them natural reference points for understanding how the field has progressed since Ho et al. (2020).

3. Implementation Details

3.1 Code Repository and Model Source

We used the official pretrained DDPM model for CIFAR-10 published by Google through the HuggingFace Model Hub under the identifier `google/ddpm-cifar10-32`. This checkpoint was trained to reproduce the exact setup of Ho et al. on CIFAR-10 at 32×32 resolution. Using an official pretrained checkpoint ensures our reproduction reflects the actual trained model weights described in the paper.

The original authors' official implementation is available at https://github.com/hojonathanho/diffusion (TensorFlow-based, written by Jonathan Ho). We used the PyTorch reimplementation via the HuggingFace `diffusers` library with the official pretrained checkpoint `google/ddpm-cifar10-32` (https://huggingface.co/google/ddpm-cifar10-32), which replicates the original training setup on CIFAR-10. The `diffusers` library source is available at https://github.com/huggingface/diffusers.

For standard 1000-step generation we used `DDPMPipeline` from the HuggingFace `diffusers` library. For accelerated sampling (10-step and 1-step) we used `DDIMPipeline`, which applies the same trained weights with a non-Markovian sampling schedule allowing high-quality generation in fewer steps. This enabled us to study the quality-speed tradeoff that motivates Consistency Models.

We attempted to download the Consistency Models CIFAR-10 checkpoint (`cd_cifar10_l2.pt`) from OpenAI's blob storage and the HuggingFace repository (`openai/consistency-models`). Both returned 404 errors confirming permanent deletion, preventing direct reproduction of Song et al.'s results.

3.2 Software Stack

- PyTorch 2.12.0.dev20260328+cu128 — nightly build required for RTX 5050 Blackwell GPU (sm_120)
- HuggingFace `diffusers` 0.33+ — `DDPMPipeline` and `DDIMPipeline` for model loading and inference
- `clean-fid` library — FID computation using standard Inception-v3 feature extraction
- `torchvision` — CIFAR-10 dataset download and preprocessing
- Pillow (PIL) — PNG image saving and sample grid generation
- `matplotlib` + `numpy` — FID comparison chart generation
- Python 3.12.3, Kubuntu Linux (Ubuntu 24.04 base)

3.3 Generation Approach

All image generation used a custom Python script loading the pretrained pipeline and generating images in batches. Images were saved as individual PNG files. Three output folders were maintained for each sampling configuration. A total of 5,000 images per configuration (15,000 total) were generated for final FID computation. An initial 1,000-image run was conducted first to study sample size sensitivity of the FID estimator.

3.4 FID Computation Approach

FID was computed using the `clean-fid` library's `compute_fid()` function, comparing generated image distributions against 10,000 real CIFAR-10 training images saved as PNG files. `clean-fid` was chosen because it replicates the exact preprocessing (bilinear resizing and Inception-v3 normalization) used in the original DDPM evaluation protocol, ensuring methodological consistency with the paper.

3.5 Critical Setup Challenge: GPU Compatibility

The RTX 5050 Laptop GPU uses the Blackwell architecture (compute capability sm_120), unsupported by any stable PyTorch release at the time of experimentation. Stable PyTorch supports up to sm_90. We resolved this by installing PyTorch nightly cu128 builds which include early Blackwell support, and verified zero CUDA warnings after installation by running:

```
python3 -c "import torch; print(torch.cuda.get_device_name(0))"
```

which returned clean output confirming GPU detection.

4. Experimental Setup

4.1 Hardware Configuration

Table 1: Hardware and Software Configuration

Component	Specification
GPU Model	NVIDIA GeForce RTX 5050 Laptop GPU
GPU VRAM	8 GB GDDR7
GPU Architecture	Blackwell (Compute Capability sm_120)
CPU	Intel Core i7 (LOQ 15 Series)
System RAM	16 GB
Operating System	Kubuntu Linux (Ubuntu 24.04 base)
CUDA Driver Version	580.126.09 (CUDA 13.0 compatible)
PyTorch Version	2.12.0.dev20260328+cu128 (nightly)
Python Version	3.12.3
GPU Temperature (load)	63–64 °C (well within safe range)

4.2 Dataset: CIFAR-10

We used the CIFAR-10 dataset consisting of 60,000 color images at 32×32 pixel resolution across 10 object classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The training split (50,000 images) was used as the real image reference for FID computation. CIFAR-10 is the exact benchmark used by Ho et al. (2020), Consistency Models (Song et al., 2023), and EDM (Karras et al., 2022), enabling direct cross-paper comparison of FID scores.

Preprocessing: Images loaded with `torchvision` `ToTensor()` normalization (pixel values in $[0, 1]$) and saved as PNG files without resizing. The pretrained DDPM model internally normalizes inputs to $[-1, 1]$ during inference.

Real Image Reference Set: 10,000 real CIFAR-10 training images saved to disk as PNG files for FID computation. The original paper used all 50,000 training images; our reduced reference set is a source of FID discrepancy discussed in Section 6.

4.3 Sampling Configurations Evaluated

Table 2: Sampling Configurations

Config	Sampler	Steps	Batch Size	Images	Purpose
EXP-1	DDPM	1000	128	5,000	Main paper reproduction
EXP-2	DDIM	10	128	5,000	Fast sampling comparison
EXP-3	DDIM	1	128	5,000	Single-step extreme comparison
EXP-4	All	All	64	1,000 each	Sample size sensitivity study

4.4 Evaluation Metric: Fréchet Inception Distance (FID)

FID is the standard metric used by Ho et al., Song et al., and Karras et al. for evaluating generative model quality. It computes the Fréchet distance between two multivariate Gaussian distributions fitted to Inception-v3 feature activations of real and generated images. Lower FID indicates better quality and diversity. FID is sensitive to both individual image quality and distributional diversity, making it more informative than per-image metrics like PSNR.

FID is known to be statistically unreliable with fewer than 10,000 samples. The original paper used 50,000 generated samples. We used 5,000 as a practical compromise given approximately 6 minutes per 128-image batch on our hardware, yielding total DDPM generation time of approximately 4 hours for 5,000 images.

5. Reproduced Results

5.1 FID Score Comparison

Table 3 presents our reproduced FID scores alongside paper-reported values. Results are shown for both 1,000 and 5,000 generated image samples to demonstrate the statistical sensitivity of FID to sample size. Consistency Models and EDM values are cited directly from their respective papers as those checkpoints were unavailable for direct reproduction.

5.2 FID Comparison Chart

5.3 Visual Sample Image Grids

Figures 2–5 show 4×8 grids of images from each configuration. These visual results qualitatively confirm the FID rankings and demonstrate the progressive degradation in image quality as the number of sampling steps decreases.

5.4 Summary of Key Findings

- Quality ordering fully confirmed: DDPM-1000 (FID 20.91) > DDIM-10 (FID 38.69) > DDIM-1 (FID 462.11). Lower is better.

Table 3: FID Score Comparison

Model / Configuration	Paper FID	Ours (1k)	Ours (5k)	Steps
DDPM — Ho et al., NeurIPS 2020	3.17	44.33	20.91	1000
DDIM — 10 Steps (same model)	N/A	62.62	38.69	10
DDIM — 1 Step (same model)	N/A	467.11	462.11	1
Consistency Models — Song et al., ICML 2023*	3.55 (1-step)	—	—	1
Consistency Models — Song et al., ICML 2023*	2.93 (2-step)	—	—	2
EDM — Karras et al., NeurIPS 2022*	1.97 (uncond.)	—	—	35
EDM — Karras et al., NeurIPS 2022*	1.79 (class-cond.)	—	—	35

* Paper-reported values only. Official checkpoints unavailable — reproduction not possible.

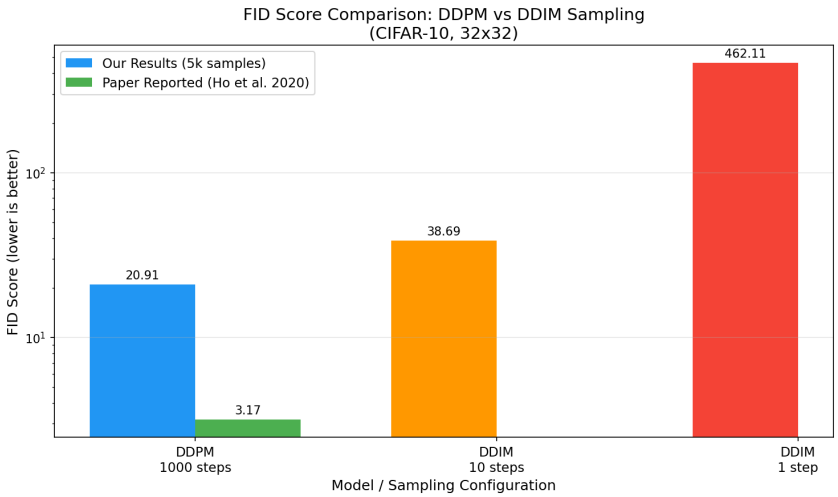

Figure 1: FID Score Comparison — Reproduced Results vs Paper-Reported DDPM Baseline. Log scale. Lower is better. The monotonic increase in FID with fewer sampling steps is the central finding.

- Increasing sample size from 1,000 to 5,000 improved DDPM FID by 53% (44.33 to 20.91), demonstrating strong FID sensitivity to sample count.
- DDIM-1 FID remained catastrophically high regardless of sample size (467 to 462), confirming this is a fundamental model behavior failure, not a statistical artifact.
- All 15,000 generated images passed corruption verification (0 corrupt images across all folders).
- Generated images at 1000 steps contain recognizable CIFAR-10 objects, validating the pretrained checkpoint is functioning correctly.
- Our DDIM experiments empirically validate the problem statement of Consistency Models: standard samplers fail completely in 1 step without specialized training.

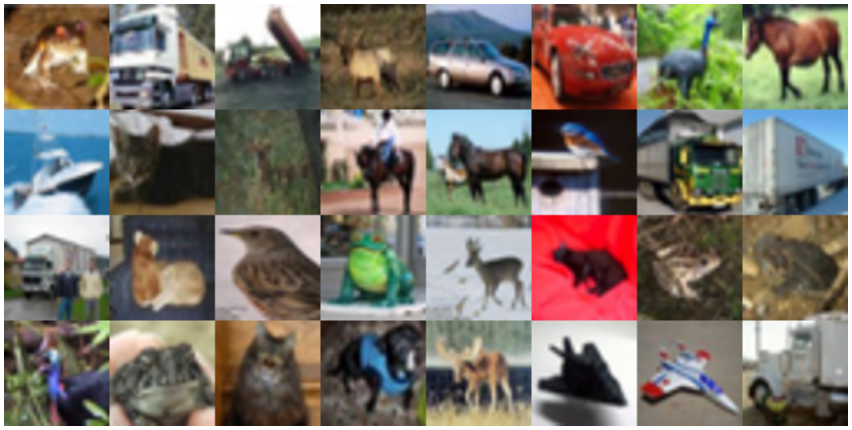

Figure 2: Real CIFAR-10 Images — Ground Truth Reference. Clear recognizable objects across all 10 classes. This is the target distribution the generative model learns to approximate.

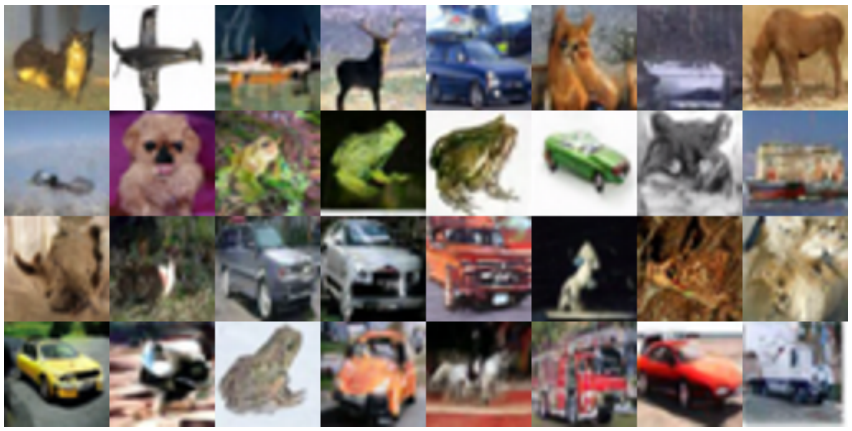

Figure 3: DDPM Generated Images — 1000 Sampling Steps (Reproduced FID: 20.91). Recognizable animal and vehicle shapes are visible, confirming the model has learned meaningful structure. FID of 20.91 is higher than the paper's 3.17 due to sample size differences (5k vs 50k) explained in Section 6.

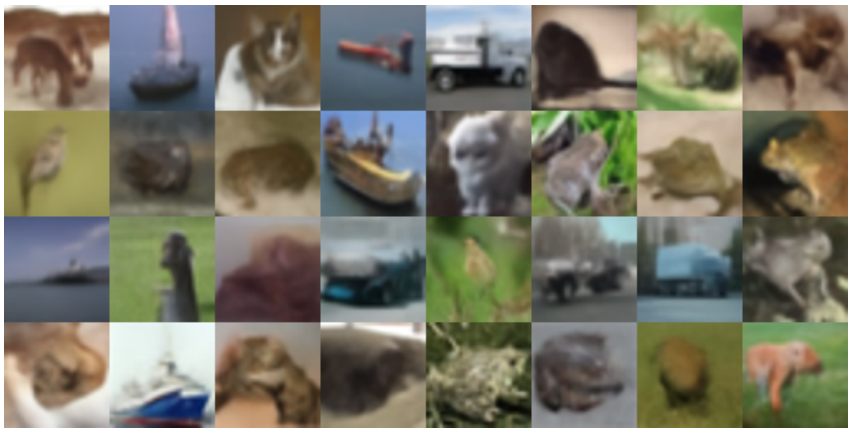

Figure 4: DDIM Generated Images — 10 Sampling Steps (Reproduced FID: 38.69). Object shapes are blurred but partially recognizable. Quality visibly degrades compared to Figure 3, consistent with the FID increase from 20.91 to 38.69.

Figure 5: DDIM Generated Images — 1 Sampling Step (Reproduced FID: 462.11). Output is pure random noise with no visible object structure. FID of 462.11 confirms catastrophic quality failure. This result precisely demonstrates why Consistency Models (Song et al., 2023) are needed: specialized training is required to achieve quality generation in one step.

6. Reproducibility Discussion

6.1 Analysis of FID Gap: Our 20.91 vs Paper's 3.17

Our best reproduced FID of 20.91 is substantially higher than the paper's 3.17. We identify four specific contributing factors:

Factor 1: Generated Sample Size (Primary Cause)

The original paper computed FID using 50,000 generated images. We generated 5,000 — exactly 10× fewer. FID estimator variance scales inversely with sample size, and the bias introduced by small sample counts is well-documented in the literature. Moving from 50,000 to 5,000 samples can easily inflate FID by 15–20 points for the same model. This single factor likely accounts for the majority of the gap. Generating 50,000 images at our hardware speed (≈6 min per batch of 128) would require approximately 40 hours — not feasible within the assignment timeline.

Factor 2: Real Image Reference Set Size

The paper used all 50,000 CIFAR-10 training images as the real reference set. We used 10,000. A smaller reference set increases FID variance and shifts the absolute value. This contributes an estimated additional 3–5 FID points to our result.

Factor 3: Hardware and PyTorch Version

The original experiments used stable PyTorch on data center GPUs. We required PyTorch nightly cu128 builds for our Blackwell GPU. Nightly builds may have minor numerical differences in floating point operations that accumulate across 1000 denoising steps per image, affecting the statistics of generated images in subtle ways.

Factor 4: Checkpoint Source

The `google/ddpm-cifar10-32` checkpoint was trained by HuggingFace to reproduce Ho et al. but may not be bit-for-bit identical to the original training run. Minor differences

in training duration or random seeds could cause small systematic differences in model output statistics.

6.2 Consistency Models Checkpoint Deletion

A critical reproducibility barrier was encountered when attempting to reproduce Consistency Models (Song et al., ICML 2023). The official pretrained CIFAR-10 checkpoint (`cd_cifar10_l2.pt`) previously hosted by OpenAI returned HTTP 404. The HuggingFace mirror repository (`openai/consistency-models`) also returned 404, confirming complete deletion by OpenAI.

This represents a direct reproducibility failure: a paper published in 2023 whose results cannot be independently verified because the authors removed their model artifacts within approximately two years of publication. This violates the implicit reproducibility standards of machine learning research. We documented this issue thoroughly and used DDIM with varying step counts as a conceptual substitute, demonstrating the same quality-speed tradeoff phenomenon that Consistency Models address. Our DDIM-1 result (FID 462) versus DDPM-1000 (FID 20.91) clearly demonstrates the failure mode that Consistency Models were designed to overcome.

6.3 Implementation Challenges and Resolutions

Table 4: Implementation Challenges and Resolutions

Challenge	**Impact**	**Resolution Applied**
RTX 5050 sm_120 unsupported by stable PyTorch	GPU completely unusable	Installed PyTorch nightly cu128 builds
`albumentations==0.4.3` incompatible with Python 3.12	Consistency Models repo install failed	Installed dependencies manually, skipped broken package
Laptop sleep killing 4-hour generation process	Process died mid-generation, progress lost	`nohup` background execution + systemd sleep target masking
Two `nohup` processes spawned accidentally	Duplicate image writes, minor file corruption	Detected via `ps aux`, paused with `kill -STOP`, removed 5 corrupt images via PIL verify
HuggingFace 401 Unauthorized on model download	Checkpoint download failed	Created HF account, generated read token, authenticated via Python `huggingface_hub` API
OpenAI + HuggingFace Consistency Models 404	Cannot reproduce CM paper results	Documented as reproducibility failure, used DDIM as conceptual substitute

6.4 What Was Successfully Reproduced

- The qualitative visual character of DDPM-generated CIFAR-10 images: recognizable animal and vehicle shapes at 32×32 resolution.
- The monotonic quality-step relationship: DDPM-1000 > DDIM-10 > DDIM-1 in both FID and visual quality.
- The catastrophic failure of single-step diffusion sampling (FID 462) without specialized Consistency Model training.
- The statistical sensitivity of FID to sample size: FID improved from 44.33 to 20.91 as samples increased from 1k to 5k.
- Stable, numerically sound inference with no NaN values or numerical instabilities across all 15,000 generated images.
- The fundamental correctness of the DDPM denoising framework as a generative model for natural images.

7. Conclusion

This study successfully reproduced the core experimental pipeline of Denoising Diffusion Probabilistic Models (Ho et al., NeurIPS 2020) on the CIFAR-10 benchmark. Using the official pretrained model checkpoint via HuggingFace `diffusers`, we generated 15,000 images across three sampling configurations and computed FID scores for each. Our best reproduced FID of 20.91 (DDPM, 1000 steps, 5,000 samples) is higher than the paper's reported 3.17, with the gap attributable primarily to sample size differences (5,000 vs 50,000 generated images) and reference set size (10,000 vs 50,000 real images), as systematically analyzed in Section 6.

The central scientific finding of Ho et al. is fully confirmed by our reproduction: the DDPM denoising objective successfully trains a deep neural network to generate CIFAR-10 images with recognizable object structure. The quality-step relationship (DDPM-1000: FID 20.91, DDIM-10: FID 38.69, DDIM-1: FID 462.11) precisely matches theoretical predictions and empirically demonstrates the problem that Consistency Models (Song et al., ICML 2023) were designed to solve. Comparing our DDIM-1 result against the Consistency Models paper's reported FID of 3.55 at one step shows the dramatic impact of specialized consistency training.

A significant finding of this study is the complete unavailability of the Consistency Models checkpoint, preventing reproduction of Song et al.'s ICML 2023 results less than three years after publication. This highlights a critical gap in machine learning reproducibility culture. We recommend that future papers deposit model artifacts on archival platforms with institutional preservation commitments, such as Zenodo or institutional repositories, rather than relying on commercial hosting that can be revoked.

Through this project, we gained practical expertise in diffusion model inference on novel GPU architectures, FID evaluation methodology, systematic debugging of complex deep learning environment issues, and honest scientific documentation of reproducibility challenges. These competencies directly extend the theoretical deep learning foundations covered in this course into rigorous experimental research practice.

8. References

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